Lecture 4: Fundamental Postulate, States, Phase Rule, and Pure Substance Phase Behavior

CBE 20260 / Thermodynamics I

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Table of contents

1 Last Class

- Forms of work
- Internal energy

- Maxwell-Boltzmann distribution
- Heat

1.1 What you should be able to do after this lecture

By the end of class you should be able to:

1. State the **fundamental postulate** of classical thermodynamics and explain what it implies for **state specification** of a pure substance.
2. Define **equilibrium** and distinguish **state variables** from **path-dependent quantities**.
3. Explain why **differences of state variables** depend only on the end states, not the path.
4. Use the **Gibbs phase rule** to determine degrees of freedom for single-phase and multiphase regions (including the triple point).
5. Interpret **P–T** and **P–V** diagrams for a pure substance, including **saturation lines**, **critical point**, **triple point**, and **normal boiling point**.
6. Describe how the **3D P–V–T surface** unifies the common 2D projections.

Brief review - Heat and Work

- Heat (Q) and Work (W) are NOT energy
- Heat and Work are mechanisms to change the energy of a system
- Energy is kinetic, potential, and internal
- Work is the "orderly" transfer of energy. It has direction and typically requires some device to facilitate it
- Heat is "disorderly" transfer of energy. It is spontaneous given a temperature difference and results from random molecular collisions.
- Heat and Work are therefore not equivalent.
- Internal energy is molecular kinetic and potential energy
- Kinetic: Temperature
- Potential: "molecular position" => more on this later

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2 Fundamental postulate of thermodynamics

The **fundamental postulate** underlies all of equilibrium thermodynamics:

> **Fundamental postulate (classical thermo):**
> *Isolated systems with no external forces acting on them will approach unique equilibrium states characterized by the mass of each distinct species in the system and two independently variable properties.*

For a **pure substance** (single chemical component), this has a powerful implication:

> If you specify **two independent intensive properties** (e.g., T and P, or T and $\underline{V}$, or P and $\hat{V}$, etc.), then **all other intensive properties are determined**.

2.1 State variables and equilibrium

A **state variable** (state function) is a property determined solely by the equilibrium state of the system. Examples include:

- T, P, $\hat{V}$ (specific volume), ρ (density)
- $\underline{U}$ (molar internal energy), $\hat{H}$ (specific enthalpy), $\underline{S}$ (molar entropy) (*later*)

For a pure substance, we can write relationships like:

$$\underline{U} = \underline{U}(T, P), \quad \underline{V} = \underline{V}(T, P), \quad \underline{H} = \underline{H}(T, P), \text{ etc.}$$

Because state variables describe equilibrium states, they have **no memory** of how the system got there.

The Fundamental Postulate means that we could also have relationships like

$$\underline{V} = \underline{V}(T, \underline{U}), \quad P = P(T, \underline{V}), \quad \underline{U} = \underline{U}(T, \underline{H}), \text{ etc.}$$

2.2 Differences in state variables do not depend on the path

Consider two equilibrium states, 1 and 2. For any state variable X,

$$\Delta X = X_2 - X_1$$

That is, differences in state variables on depend on the endpoints. This is one of the most practical ideas in thermodynamics:

> Even if the **real process path** from state 1 to state 2 is complicated, we can compute ΔX using **any convenient (hypothetical) path**, as long as it connects the same endpoints.

2.2.1 Convenient thermodynamic paths

A common strategy is to split a change into two steps—e.g., first change P at constant T, then change T at constant P (or vice versa), as shown in Fig. 1

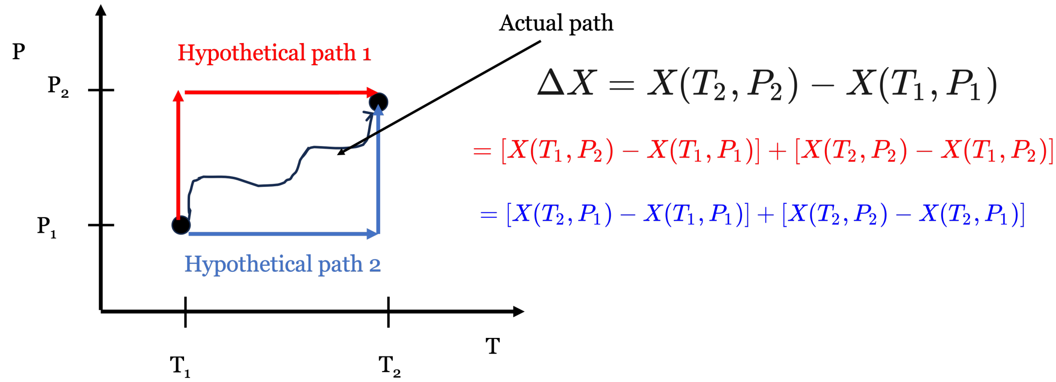

Fig. 1: Different paths connecting state variables at two conditions.

$$\Delta X = X(T_2, P_2) - X(T_1, P_1)$$

One convenient decomposition is:

$$\Delta X = \underbrace{X(T_1, P_2) - X(T_1, P_1)}_{\text{step 1: change } P \text{ at fixed } T_1} + \underbrace{X(T_2, P_2) - X(T_1, P_2)}_{\text{step 2: change } T \text{ at fixed } P_2}$$

Or alternatively:

$$\Delta X = \underbrace{X(T_2, P_1) - X(T_1, P_1)}_{\text{step 1: change } T \text{ at fixed } P_1} + \underbrace{X(T_2, P_2) - X(T_2, P_1)}_{\text{step 2: change } P \text{ at fixed } T_2}$$

Key point: A *real* path might involve simultaneous changes in T and P in a complicated way, but for a state variable we are free to choose a **thermodynamic path** that is easier to evaluate.

2.2.2 Path-dependent quantities (contrast)

Not everything is a state variable. The most important counterexamples are:

- heat Q
- work W

These depend on the process details (path), not just end states. (We will make this precise as we proceed.) But for now, remember that **heat and work are NOT state variables**; they depend on path or the actual thermodynamic process.

2.3 Gibbs phase rule

The fundamental postulate suggests that we need only specify two independent quantities for a pure substance, and all other state variables are set. This is a remarkable statement. Although macroscopic substances are made up of more than 10^{23} atoms, we need only specify two variables like T and P and **all properties** are fixed. You have seen this before when looking up properties in steam tables (Fig. 2). You can look up the density (inverse specific volume) or internal energy of water at a given T and P when water is either subcooled or superheated.

TABLE B.4 Superheated Water Vapor 10

$P = 10$ kPa

T °C	$\hat{v}$ m³/kg	$\hat{u}$ kJ/kg	$\hat{h}$ kJ/kg	$\hat{s}$ kJ/kg K
sat	14.674	2437.9	2584.6	8.1501
50	14.869	2443.9	2592.6	8.1749
100	17.196	2515.5	2687.5	8.4479
150	19.513	2587.9	2783.0	8.6881
200	21.825	2661.3	2879.5	8.9037
250	24.136	2736.0	2977.3	9.1002
300	26.445	2812.1	3076.5	9.2812
400	31.063	2968.9	3279.5	9.6076
500	35.679	3132.3	3489.0	9.8977
600	40.295	3302.5	3705.4	10.1608
700	44.911	3479.6	3928.7	10.4028
800	49.526	3663.8	4159.1	10.6281
900	54.141	3855.0	4396.4	10.8395
1000	58.757	4053.0	4640.6	11.0392
1100	63.372	4257.5	4891.2	11.2287
1200	67.987	4467.9	5147.8	11.4090
1300	72.603	4683.7	5409.7	11.5810

$P = 50$ kPa

T °C	$\hat{v}$ m³/kg	$\hat{u}$ kJ/kg	$\hat{h}$ kJ/kg	$\hat{s}$ kJ/kg K
sat	3.240	2483.8	2645.9	7.5939
100	3.418	2511.6	2682.5	7.6947
150	3.889	2585.6	2780.1	7.9400
200	4.356	2659.8	2877.6	8.1579
250	4.821	2735.0	2976.0	8.3555
300	5.284	2811.3	3075.5	8.5372
400	6.209	2968.4	3278.9	8.8641
500	7.134	3131.9	3488.6	9.1545
600	8.058	3302.2	3705.1	9.4177
700	8.981	3479.5	3928.5	9.6599
800	9.904	3663.7	4158.9	9.8852
900	10.828	3854.9	4396.3	10.0967
1000	11.751	4052.9	4640.5	10.2964
1100	12.674	4257.4	4891.1	10.4858
1200	13.597	4467.8	5147.7	10.6662
1300	14.521	4683.6	5409.6	10.8382

$P = 100$ kPa

T °C	$\hat{v}$ m³/kg	$\hat{u}$ kJ/kg	$\hat{h}$ kJ/kg	$\hat{s}$ kJ/kg K
sat	1.6940	2506.1	2675.5	7.3593
100	1.6958	2506.6	2676.2	7.3614
150	1.9364	2582.7	2776.4	7.6133
200	2.1723	2658.0	2875.3	7.8342
250	2.4060	2733.7	2974.3	8.0332
300	2.6388	2810.4	3074.3	8.2157
400	3.1026	2967.8	3278.1	8.5434
500	3.5655	3131.5	3488.1	8.8341
600	4.0278	3301.9	3704.7	9.0975
700	4.4899	3479.2	3928.2	9.3398
800	4.9517	3663.5	4158.7	9.5652
900	5.4135	3854.8	4396.1	9.7767
1000	5.8753	4052.8	4640.3	9.9764
1100	6.3370	4257.3	4890.9	10.1658
1200	6.7986	4467.7	5147.6	10.3462
1300	7.2603	4683.5	5409.5	10.5182

$P = 200$ kPa

T °C	$\hat{v}$ m³/kg	$\hat{u}$ kJ/kg	$\hat{h}$ kJ/kg	$\hat{s}$ kJ/kg K
sat	0.88573	2529.5	2706.6	7.1271
150	0.95964	2576.9	2768.8	7.2795
200	1.08034	2654.4	2870.5	7.5066
250	1.19880	2731.2	2971.0	7.7085
300	1.31616	2808.6	3071.8	7.8926
400	1.54930	2966.7	3276.5	8.2217
500	1.78139	3130.7	3487.0	8.5132
600	2.01297	3301.4	3704.0	8.7769
700	2.24426	3478.8	3927.7	9.0194
800	2.47539	3663.2	4158.3	9.2450
900	2.70643	3854.5	4395.8	9.4565
1000	2.93740	4052.5	4640.0	9.6563
1100	3.16834	4257.0	4890.7	9.8458
1200	3.39927	4467.5	5147.3	10.0262
1300	3.63018	4683.2	5409.3	10.1982

$P = 300$ kPa

T °C	$\hat{v}$ m³/kg	$\hat{u}$ kJ/kg	$\hat{h}$ kJ/kg	$\hat{s}$ kJ/kg K
sat	0.60582	2543.6	2725.3	6.9918
150	0.63388	2570.8	2761.0	7.0778
200	0.71629	2650.7	2865.5	7.3115
250	0.79636	2728.7	2967.6	7.5165
300	0.87529	2806.7	3069.3	7.7022
400	1.03151	2965.5	3275.0	8.0329
500	1.18669	3130.0	3486.0	8.3250
600	1.34136	3300.8	3703.2	8.5892
700	1.49573	3478.4	3927.1	8.8319
800	1.64994	3662.9	4157.8	9.0575
900	1.80406	3854.2	4395.4	9.2691
1000	1.95812	4052.3	4639.7	9.4689
1100	2.11214	4256.8	4890.4	9.6585
1200	2.26614	4467.2	5147.1	9.8389
1300	2.42013	4683.0	5409.0	10.0109

$P = 400$ kPa

T °C	$\hat{v}$ m³/kg	$\hat{u}$ kJ/kg	$\hat{h}$ kJ/kg	$\hat{s}$ kJ/kg K
sat	0.46246	2553.6	2738.5	6.8958
150	0.47084	2564.5	2752.8	6.9299
200	0.53422	2646.8	2860.5	7.1706
250	0.59512	2726.1	2964.2	7.3788
300	0.65484	2804.8	3066.7	7.5661
400	0.77262	2964.4	3273.4	7.8984
500	0.88934	3129.2	3484.9	8.1912
600	1.00555	3300.2	3702.4	8.4557
700	1.12147	3477.9	3926.5	8.6987
800	1.23722	3662.5	4157.4	8.9244
900	1.35288	3853.9	4395.1	9.1361
1000	1.46847	4052.0	4639.4	9.3360
1100	1.58404	4256.5	4890.1	9.5255
1200	1.69958	4467.0	5146.8	9.7059
1300	1.81511	4682.8	5408.8	9.8780

Fig. 2: Snapshot from a portion of the superheated steam tables.

At the same T, properties change with changing P. Likewise, at the same P, properties change with changing T.

Something different happens when you look at the *saturated* steam tables (Fig. 3)

STEAM TABLES Saturated Water - Temperature Table										
Temp. °C T	Sat. Press. kPa p_{sat}	Specific Volume m^3/kg		Internal Energy kJ/kg			Enthalpy kJ/kg		
		Sat. liquid v_f	Sat. vapor v_g	Sat. liquid u_f	Evap. u_{fg}	Sat. vapor u_g	Sat. liquid h_f	Evap. h_{fg}	Sat. vapor h_g
0.01	0.6113	0.001 000	206.14	0.00	2375.3	2375.3	0.01	2501.3	2501.4
5	0.8721	0.001 000	147.12	20.97	2361.3	2382.3	20.98	2489.6	2510.6
10	1.2276	0.001 000	106.38	42.00	2347.2	2389.2	42.01	2477.7	2519.8
15	1.7051	0.001 001	77.93	62.99	2333.1	2396.1	62.99	2465.9	2528.9
20	**2.339**	**0.001 002**	**57.79**	**83.95**	**2319.0**	**2402.9**	**83.96**	**2454.1**	**2538.1**
25	3.169	0.001 003	43.36	104.88	2304.9	2409.8	104.89	2442.3	2547.2
30	4.246	0.001 004	32.89	125.78	2290.8	2416.6	125.79	2430.5	2556.3
35	5.628	0.001 006	25.22	146.67	2276.7	2423.4	146.68	2418.6	2565.3
40	7.384	0.001 008	19.52	167.56	2262.6	2430.1	167.57	2406.7	2574.3
45	**9.593**	**0.001 010**	**15.26**	**188.44**	**2248.4**	**2436.8**	**188.45**	**2394.8**	**2583.2**
50	12.349	0.001 012	12.03	209.32	2234.2	2443.5	209.33	2382.7	2592.1
55	15.758	0.001 015	9.568	230.21	2219.9	2450.1	230.23	2370.7	2600.9
60	19.940	0.001 017	7.671	251.11	2205.5	2456.6	251.13	2358.5	2609.6
65	25.03	0.001 020	6.197	272.02	2191.1	2463.1	272.06	2346.2	2618.3
70	**31.19**	**0.001 023**	**5.042**	**292.95**	**2176.6**	**2569.6**	**292.98**	**2333.8**	**2626.8**
75	38.58	0.001 026	4.131	313.90	2162.0	2475.9	313.93	2321.4	2635.3
80	47.39	0.001 029	3.407	334.86	2147.4	2482.2	334.91	2308.8	2643.7
85	57.83	0.001 033	2.828	355.84	2132.6	2488.4	355.90	2296.0	2651.9
90	70.14	0.001 036	2.361	376.85	2117.7	2494.5	376.92	2283.2	2660.1
95	**84.55**	**0.001 040**	**1.982**	**397.88**	**2102.7**	**2500.6**	**397.96**	**2270.2**	**2668.1**
	MPa								
100	0.101 35	0.001 044	1.6729	418.94	2087.6	2506.5	419.04	2257.0	2676.1
105	0.120 82	0.001 048	1.4194	440.02	2072.3	2512.4	440.15	2243.7	2683.8
110	0.143 27	0.001 052	1.2102	461.14	2057.0	2518.1	461.30	2230.2	2691.5
115	0.169 06	0.001 056	1.0366	482.30	2041.4	2523.7	482.48	2216.5	2699.0
120	**0.198 53**	**0.001 060**	**0.8919**	**503.50**	**2025.8**	**2529.3**	**503.71**	**2202.6**	**2706.3**

Fig. 3: Snapshot from a portion of the saturated steam tables.

Here, we only specify one variable (i.e. T) and all the other properties are set. The pressure **must** be the saturation pressure, whereas when the system was in the superheated state, there could be multiple pressures at the same temperature. Does this violate the Fundamental Postulate? No! The requirement that the system be in phase equilibrium imposes a constraint that takes one of the independent variables away.

The **Gibbs phase rule** tells us how many independent variables can be specified for any system:

$$F = C - \Pi + 2 \tag{1}$$

where:

- F = degrees of freedom (number of independent intensive variables you can set)
- C = number of chemical components
- Π = number of phases present (solid, liquid, vapor, etc.)

2.3.1 Pure substances ($C = 1$)

For a pure substance ($C = 1$):

- **Single phase** ($\Pi = 1$):

$$F = 1 - 1 + 2 = 2$$

  Two independent intensive variables (e.g., T and P) specify the state.

- **Two phases** ($\Pi = 2$), e.g., liquid + vapor:

  $$F = 1 - 2 + 2 = 1$$

  Only one independent intensive variable can be chosen. Example: specifying T fixes the saturation pressure $P_{\text{sat}}(T)$. That is what we see in the saturated steam tables, but the same could be true of we had solid and liquid in phase equilibrium.

- **Three phases** ($\Pi = 3$), solid + liquid + vapor:

  $$F = 1 - 3 + 2 = 0$$

  No freedom: the system must be at the **triple point**, where all thermodynamic properties are specified.

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3 Physical properties of pure substances and phase behavior

We will focus heavily on **pure substances** in this course. The most accessible "primary" measurable properties are:

- pressure P
- temperature T
- molar volume $\underline{V}$ (or specific volume $\hat{V}$, or density, ρ)

Collectively, this is often described as **P–V–T behavior**.

The fundamental postulate tells us: for a pure substance in equilibrium, **two** independent intensive variables are enough to determine the state. That is why 2D projections like P–T and P–V diagrams are so useful. It is also why the steam tables are set up as they are.

3.1 P–T diagrams for a pure substance

A **P–T phase diagram** shows where solid, liquid, and vapor are stable. A typical diagram includes:

- sublimation line (solid–vapor)
- fusion/melting line (solid–liquid)
- vaporization/boiling line (liquid–vapor)
- triple point
- critical point

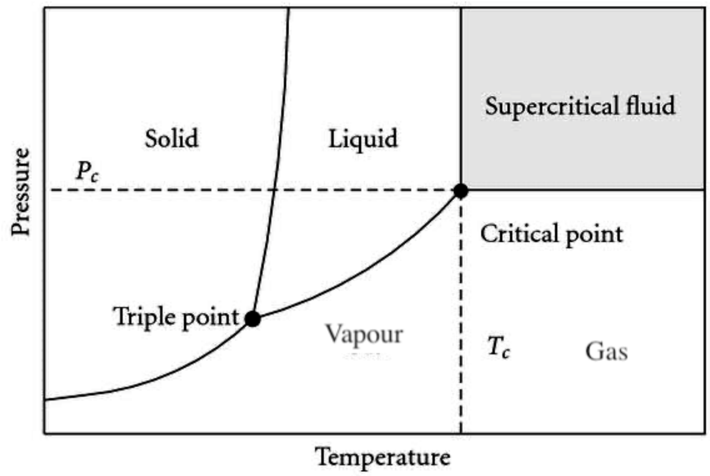

Fig. 4: Typical P–T phase diagram showing triple point, critical point, and phase regions.

3.1.1 5.1 Triple point

The **triple point** is the unique point where solid, liquid, and vapor coexist:

- $\Pi = 3, C = 1 \Rightarrow F = 0$

So T and P are both fixed at the triple point.

3.1.2 Boiling point and saturation pressure

On the liquid–vapor coexistence line:

- $\Pi = 2$, so $F = 1$.

Thus at a **given pressure** there is a **single temperature** at which boiling occurs, and at a given temperature there is a **single saturation pressure**:

$$P_{\text{sat}} = P_{\text{sat}}(T)$$

As T increases, $P_{\text{sat}}(T)$ increases.

Normal boiling point: the temperature at which liquid and vapor coexist at $P = 1$ bar. If you live in the mountains, you may see that some cooking directions are modified. That's because water boils at a lower temperature at high elevation (see Fig. 5).

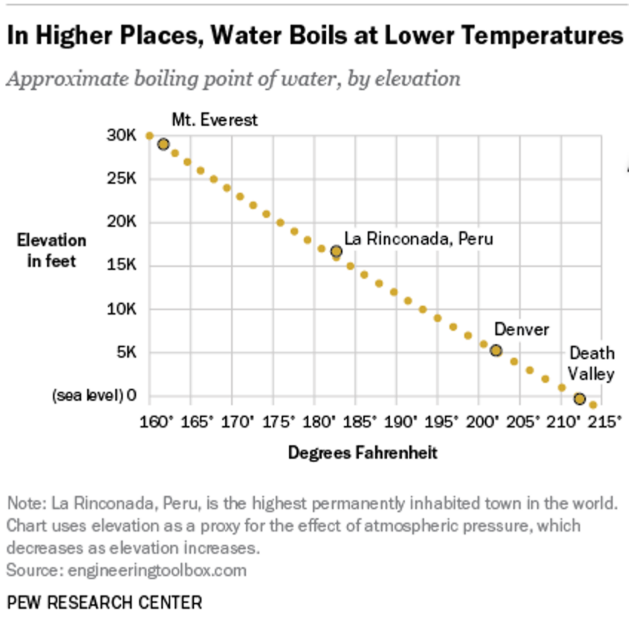

Fig. 5: Boiling point decreases with elevation because atmospheric pressure decreases.

3.1.3 Melting and sublimation lines

- **Melting line (solid–liquid):** locus of points where solid and liquid coexist.
-
- **Sublimation line (solid–vapor):** locus of points where solid and vapor coexist.

Different substances have different slopes for these lines (water is a famous exception in some respects), but the phase-rule logic is universal.

3.1.4 Critical point and supercritical fluid

The **critical point** is the unique endpoint of the liquid–vapor coexistence curve where the distinction between liquid and vapor disappears. It is characterized by:

- critical temperature T_c
- critical pressure P_c

For $T > T_c$ and $P > P_c$, the substance is a **supercritical fluid** (no liquid–vapor phase boundary).

3.2 P–V diagrams for a pure substance

A **P–V diagram** shows pressure versus (molar or specific) volume, often with **isotherms** (constant T curves).

A typical set of isotherms looks like those in Fig. 6 :

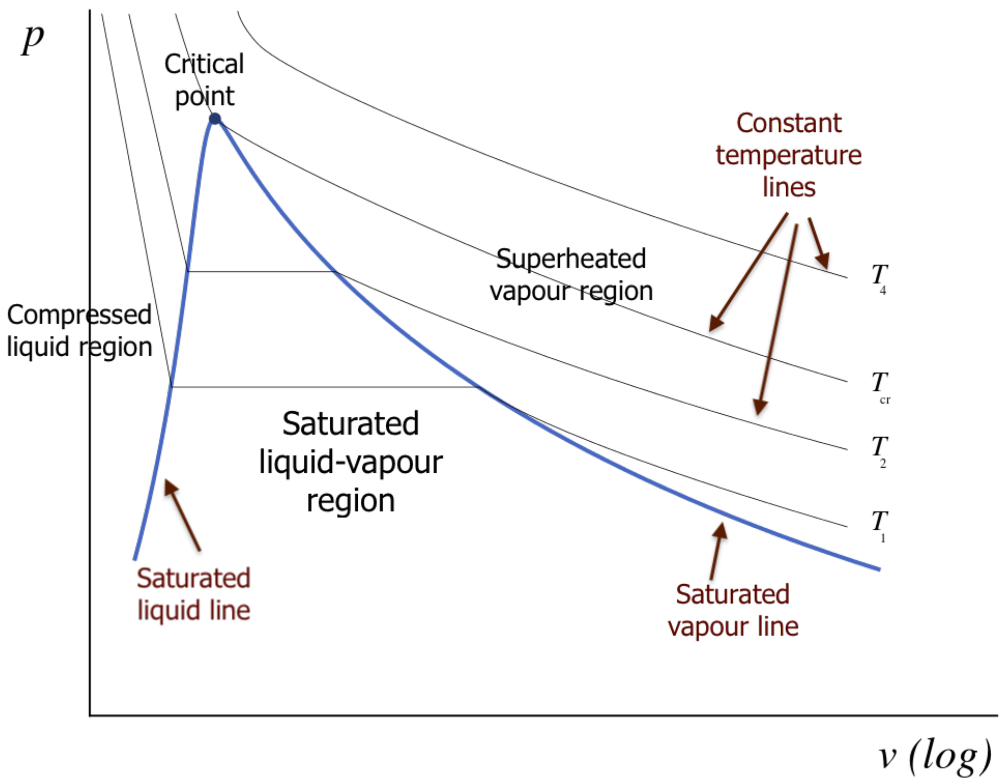

Fig. 6: Typical P–V behavior for a pure substance showing saturated liquid/vapor lines and the critical point. (Image from "Introduction to Engineering Thermodynamics", Claire Yu Yan CC-BY-NC-SA).

3.2.1 Saturated liquid and saturated vapor lines

For $T < T_c$, an isotherm intersects a two-phase region where **liquid and vapor coexist**. The boundaries are:

- **Saturated liquid line**: states that are "just about to boil" (quality $x = 0$)
-
- **Saturated vapor line**: states that are "just about to condense" (quality $x = 1$)

Together these form the **vapor dome** on many property diagrams.

Inside the two-phase region:

- pressure is fixed at $P_{\text{sat}}(T)$
- $\underline{V}$ depends on the phase fractions

3.2.2 Quality (two-phase mixtures)

For a liquid–vapor mixture, we often define the **vapor quality** x:

$$x = \frac{m_v}{m_{\text{total}}}$$

For any specific property y that is additive on a mass basis (e.g., $\underline{V}$, $\underline{U}$, $\underline{S}$),

$$y = (1 - x)\, y_\ell + x\, y_v$$

where:

- subscript ℓ denotes saturated liquid
- subscript v denotes saturated vapor

This relation becomes a practical tool once we start using property tables and charts.

3.2.3 How saturation pressure changes with temperature (reading the diagram)

A useful qualitative fact seen in the diagrams:

- At higher T (but still $T < T_c$), the saturation pressure is larger:

$$P_{\text{sat}}(T_2) > P_{\text{sat}}(T_1) \quad \text{if } T_2 > T_1$$

- At a fixed pressure, increasing temperature tends to increase the specific volume in the vapor region.

These are the kinds of trends we will use repeatedly in problem solving.

> Interactive Example
>
> I have prepared an interactive Jupyter Notebook that allows you to plot a pressure-volume diagram. It uses information from an open source data package called CoolProp. We will talk more about this later, but for now just focus on what the diagram looks like.
> *Note: This example is available as an interactive Jupyter Notebook on the course website.*

3.3 T-V diagrams for a pure substance

You can also plot T vs $\underline{V}$; Fig. 7 shows an example of what such a plot looks like for the liquid and vapor regions of a pure substance.

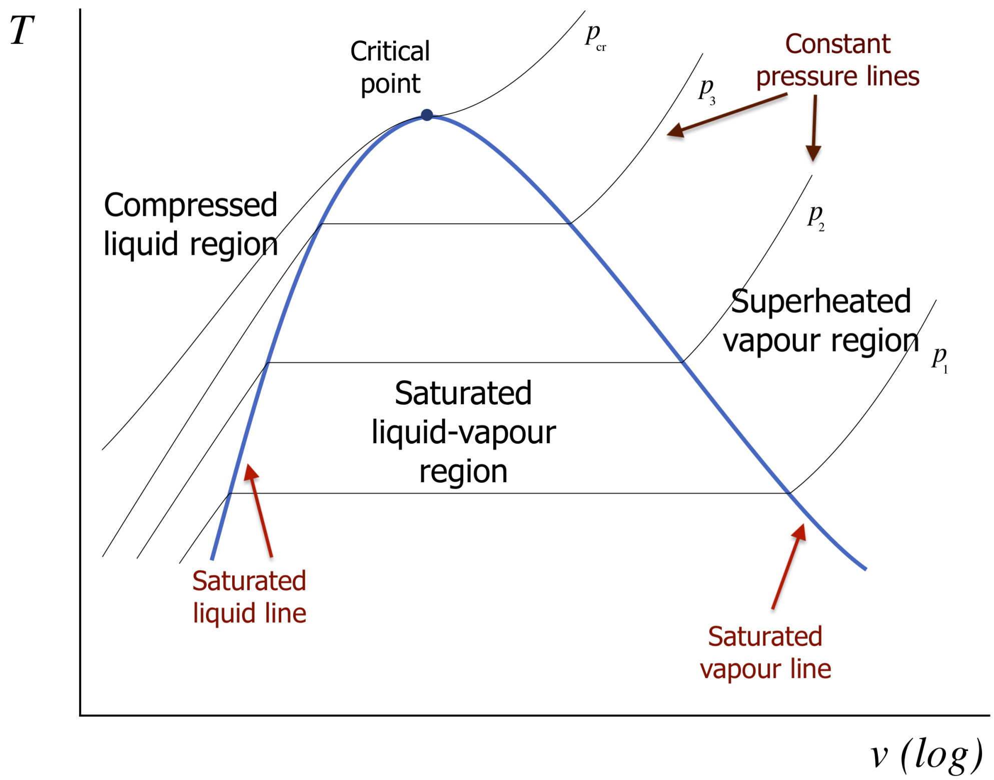

Fig. 7: Typical T–V behavior for a pure substance showing saturated liquid/vapor lines and the critical point. (Image from "Introduction to Engineering Thermodynamics", Claire Yu Yan CC-BY-NC-SA).

3.4 3-D representation: the P–V–T surface

The P–T and P–V diagrams are **projections** of a more complete picture: the **3D P–V–T surface**, as shown in Fig. 8.

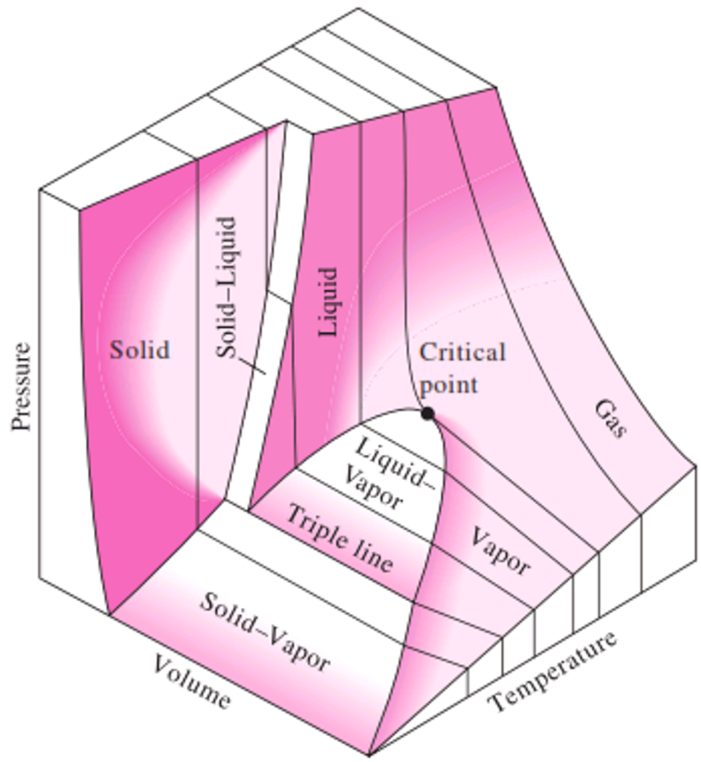

Fig. 8: Schematic 3D P–V–T surface showing solid, liquid, vapor regions, critical point, and a triple line. Image courtesy of Sounak Bhattacharjee.

Key ideas:

- The two-phase equilibria form **surfaces/lines** on this 3D object.
-
- The triple point in a P–T diagram corresponds to a **triple line** in 3D (where three phases coexist).
-
- The critical point is a special point on the liquid–vapor boundary where the two-phase region ends.

A helpful way to interpret the surface is to imagine processes such as **isobaric heating** (constant P) and observe how the state moves across single-phase regions and possibly through two-phase coexistence.

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4 Summary and what comes next

- The fundamental postulate tells us: for a pure substance, **two** independent intensive variables determine the equilibrium state.

- State variables have no path dependence: ΔX depends only on endpoints.
- The Gibbs phase rule explains degrees of freedom:
  - single phase (pure): $F = 2$
  - two phases (pure): $F = 1$
  - triple point (pure): $F = 0$
- P–T and P–V diagrams are essential tools for visualizing phase behavior.
- The P–V–T surface unifies these projections.

Next lecture: we will begin using quantitative property relations/tables/charts to compute properties and state changes for pure substances in a systematic way.